Introduction

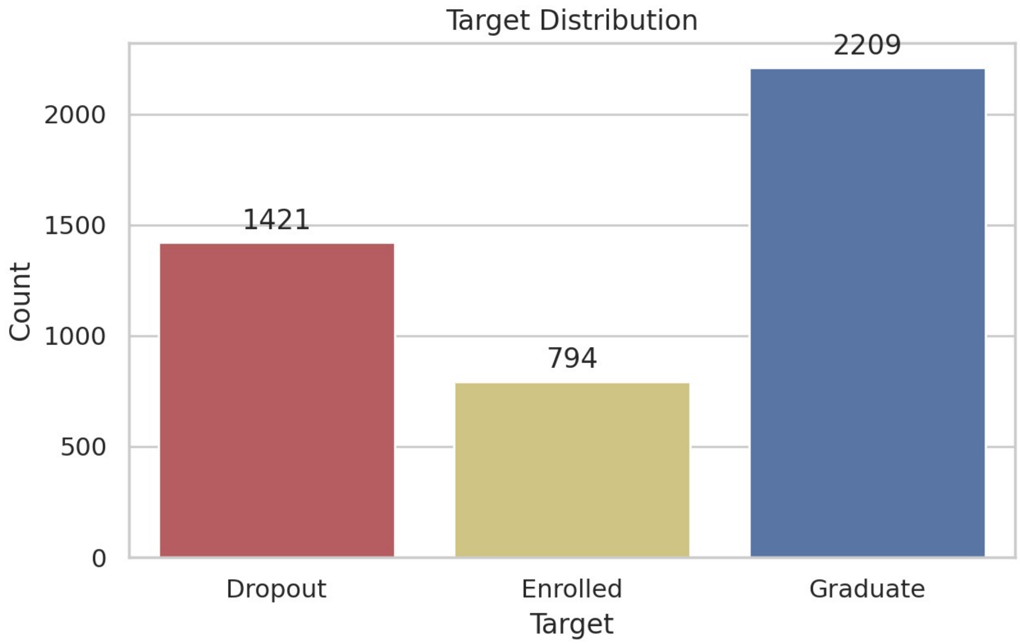
Student dropout risk analysis with reproducible ML workflow

4424 Students	**36** Raw features	**96** Processed features

Target distribution: Dropout 1421, Enrolled 794, Graduate 2209.

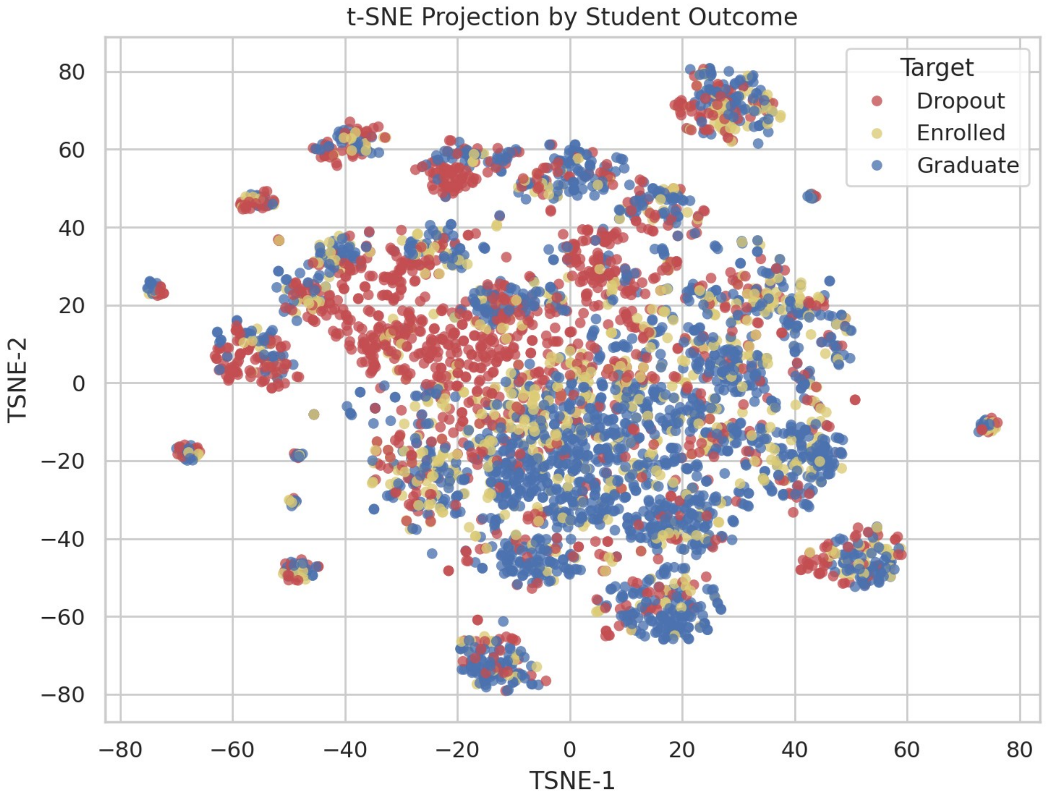
Pipeline: preprocessing, t-SNE, clustering, binary prediction, validation, open exploration.

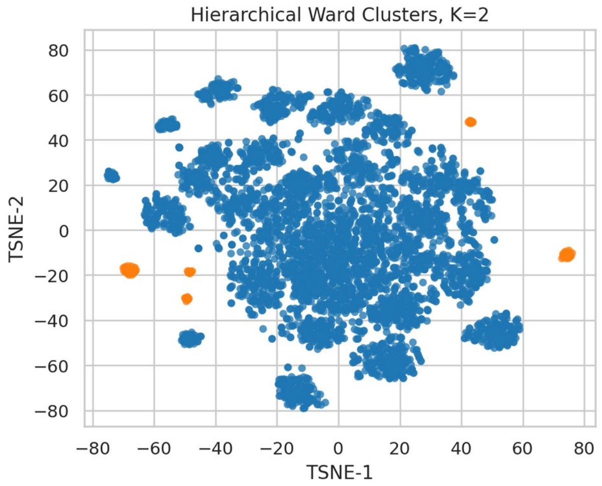
Main supervised task: detect Dropout against Graduate after excluding uncertain Enrolled labels.



t-SNE & Clustering

Local structure is visible, but true labels remain strongly overlapped





Algorithm	K	Sil.	ARI
K-Means	2	0.111	0.072
Ward	2	0.411	0.001

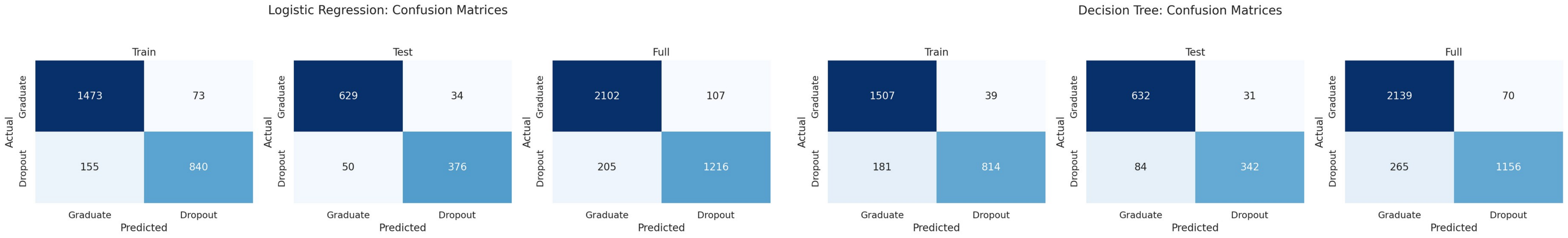
Both methods select K=2, suggesting a coarse high-risk vs lower-risk split.

Ward has stronger internal separation; K-Means aligns better with true labels by ARI.

Low ARI confirms that natural clusters are not equivalent to academic outcomes.

Prediction Results

Confusion matrices for train, test, and full binary data



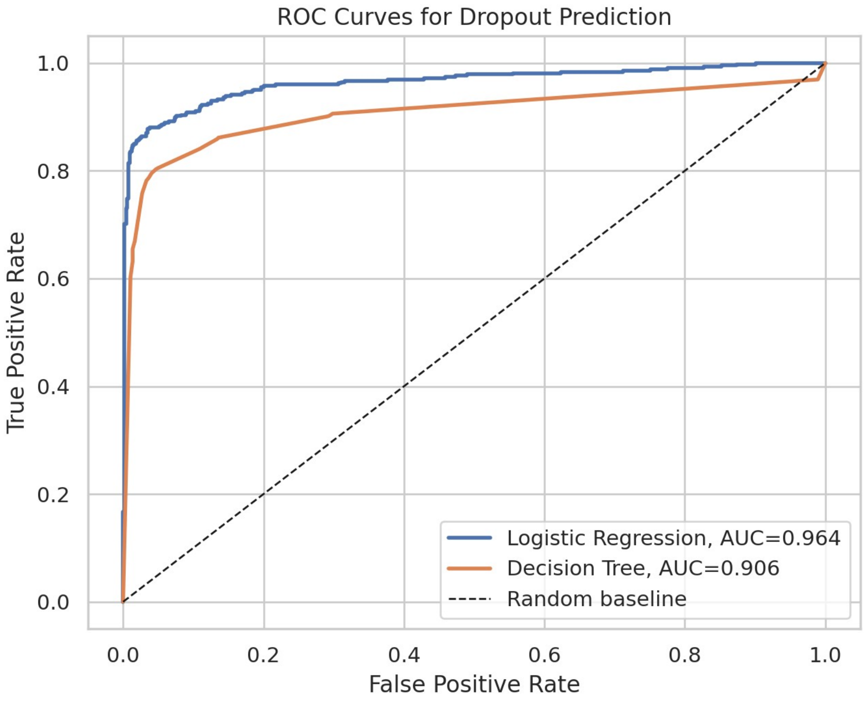
Binary target: Dropout is positive, Graduate is negative.

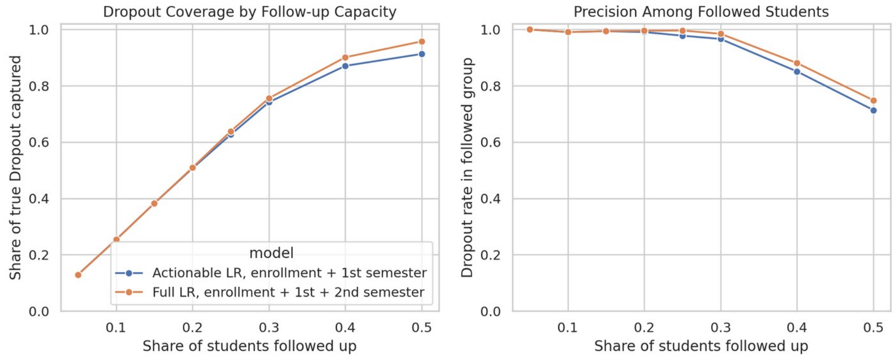
Logistic regression test matrix: 376 true Dropout detected, 50 missed.

Decision tree test matrix: 342 true Dropout detected, 84 missed.

Evaluation & Intervention

ROC/AUC plus coverage under limited advising capacity





Model	Acc.	Recall	F1	AUC
Logistic	0.923	0.883	0.900	0.964
Tree	0.894	0.803	0.856	0.906
RF tuned	0.921	0.857	0.895	0.958

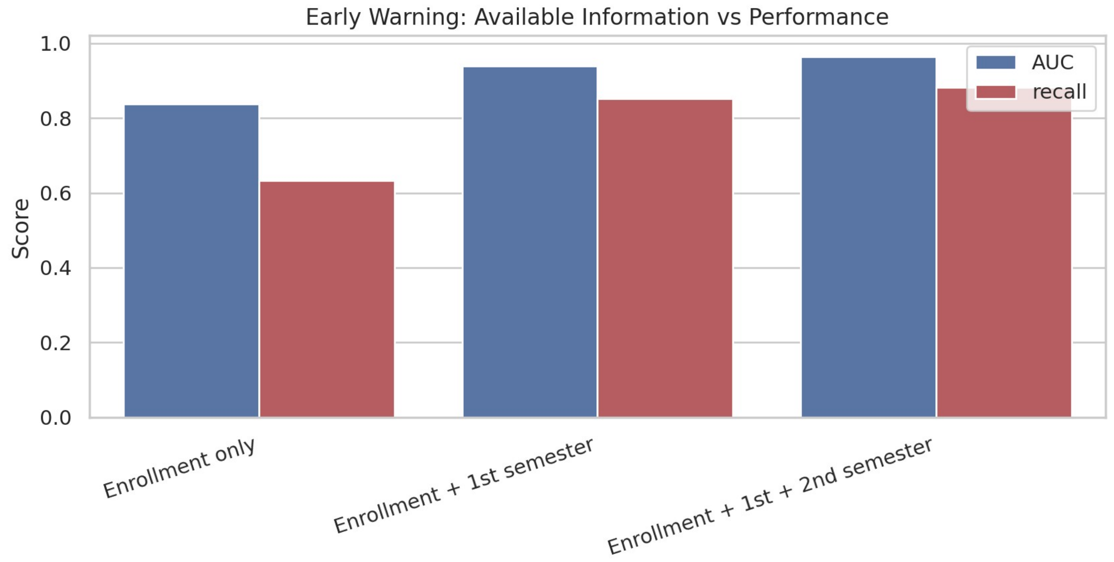
5-fold CV AUC: logistic 0.951±0.006, tree 0.914±0.004.

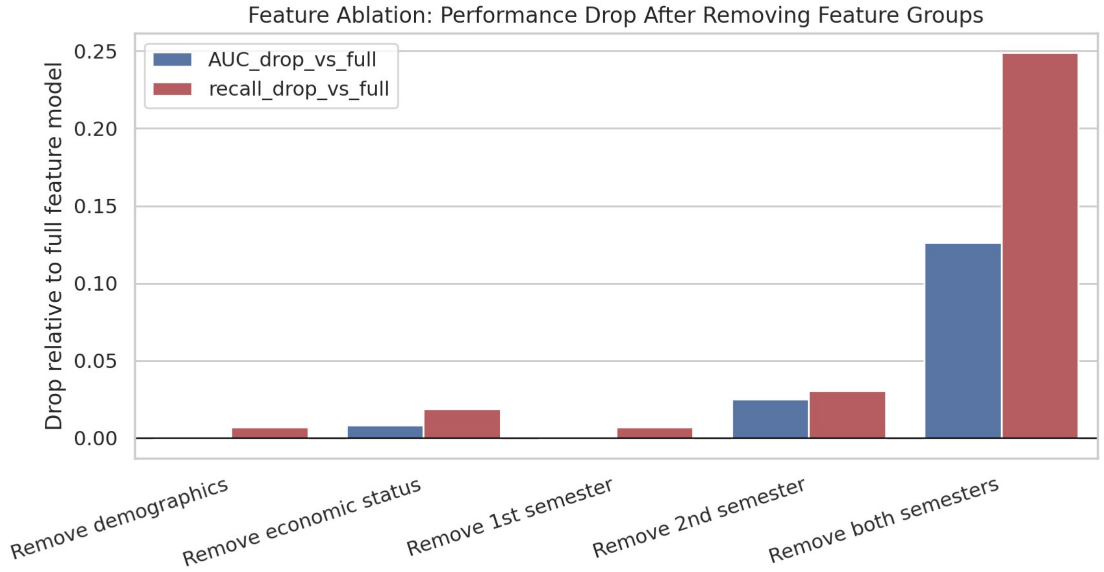
Top 25% actionable first-semester follow-up covers 0.627 of true Dropout cases.

Full second-semester model covers 0.638 at the same capacity, but is later for prevention.

Early Warning & Ablation

The first semester is the practical intervention window





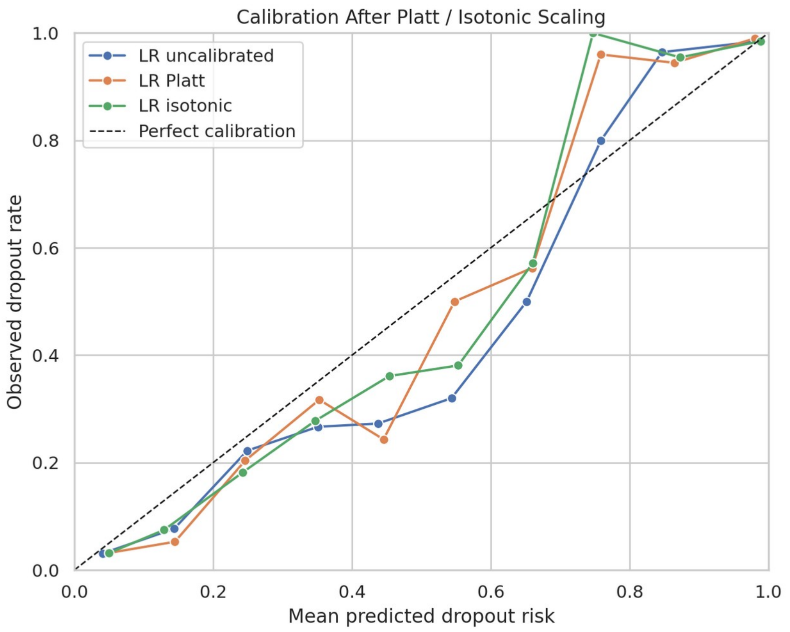
Enrollment-only AUC 0.838 rises to 0.939 after first-semester data.

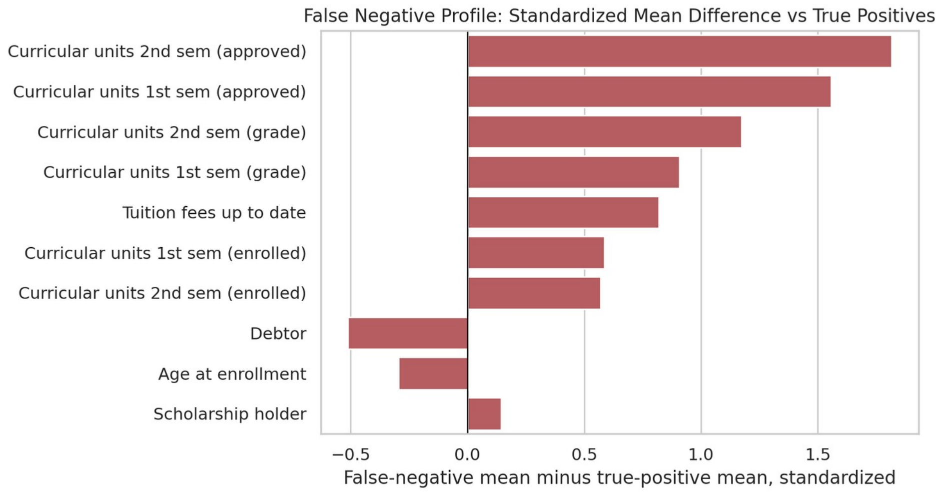
Adding second-semester data reaches AUC 0.964, useful but later for intervention.

Removing both semesters drops recall by 0.249; removing second semester drops AUC by 0.025.

Trustworthy Explanation & Risk

Calibrated probability, uncertainty, and failure-mode analysis





Deployment Checks

0.057

best Brier score

0.951-0.976

LR AUC 95% CI

50 FN

missed Dropout cases

Counterfactual-style scenario: tuition up-to-date and +3 approved units lowers predicted risk by 0.166 on average.

False negatives look stronger academically; top FN contrast is Curricular units 2nd sem (approved).

Dataset: UCI Predict Students' Dropout and Academic Success | Tools: pandas, scikit-learn, matplotlib, seaborn, imbalanced-learn